

Small Left Ventricle in Patients With Atrial Fibrillation Is Associated With Increased Cardiovascular Risk



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ABSTRACT

BACKGROUND It is still unclear whether small left ventricle (LV) is an adverse structural prognostic feature in patients with atrial fibrillation (AF).

OBJECTIVES The purpose of this study was to evaluate the association between small LV and risk of cardiovascular events in AF population.

METHODS From the China-AF registry, 7,764 patients with AF were enrolled and divided into groups with normal, small, and large LV size based on left ventricular end-diastolic dimension (LVEDD) measurement per the American Society of Echocardiography references. Cox models were used to assess the association between LV size or LVEDD with composite cardiovascular events (cardiovascular death, ischemic stroke or systemic embolism, or major bleeding).

RESULTS There were 308 (4.0%) participants assessed with small LV who were older, with lower body mass and blood pressure, and fewer comorbidities, and 429 (5.5%) were identified with large LV. Compared with the normal LV group, small LV and large LV were significantly associated with higher incidence of composite cardiovascular events (adjusted HR [aHR]: 1.54 [95% CI: 1.07-2.20] for small LV; aHR: 1.36 [95% CI: 1.02-1.81] for large LV) and cardiovascular death (aHR: 1.94 [95% CI: 1.14-3.28] for small LV; aHR: 1.83 [95% CI: 1.24-2.69] for large LV). Small LV was also associated with increased risk of major bleeding [aHR: 2.21 [95% CI: 1.01-4.86]]. A U-shaped relationship between LVEDD and composite cardiovascular events was identified ($P_{\text{nonlinear}} < 0.001$).

CONCLUSIONS In a prospective AF cohort, small LV was independently associated with an increased risk of cardiovascular events, which needed consideration in risk stratification and management for patients with AF. (ChiCTR-OCH-13003729) (J Am Coll Cardiol 2024;83:1957-1969) © 2024 by the American College of Cardiology Foundation.

Atrial fibrillation (AF) is the most common cardiac arrhythmia, imposing a severe burden on both mortality and quality of life.^{1,2} Catheter ablation has shown efficacy as first-line therapy of AF in maintaining sinus rhythm and reducing hospitalizations.³ Recent guidelines highlighted the importance of imaging evaluation of cardiac structure

and function for risk stratification in patients with AF.^{4,5} The enlarged left ventricle (LV) has been recognized as a marker of cardiac remodeling and reduced systolic function, which implies a poorer prognosis.⁶ In contrast, the manifestation and adverse effects of a small LV is largely neglected, indicative of an imbalanced focus and knowledge deficiency.



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ABBREVIATIONS AND ACRONYMS

AF = atrial fibrillation

CV = cardiovascular

DOAC = direct oral
anticoagulant

LV = left ventricle/ventricular

LVEDD = left ventricular end-
diastolic dimension

LVEF = left ventricular ejection
fraction

HF = heart failure

IS = ischemic stroke

SE = systematic embolism

The current observations have hinted a potential neurocirculatory asthenia and volume hypoperfusion in the population with a small LV.^{7,8} Although the association between small LV and cardiovascular (CV) risk remains inconclusive in the general population, the prognosis of these patients still merit attention when concomitant with hemodynamic abnormalities.^{9–11} The onset of AF induces irregular ventricular rate and reduced ventricular filling, culminating in a decreased cardiac output that exacerbates the hypoperfusion status.^{12,13} Hence, it raises concerns that whether small LV signifies an unidentified anatomical risk factor for poor prognosis in AF patients.

The China-AF registry is a prospective, multicenter, and hospital-based registry of patients with AF in China, which collected detailed data on baseline characteristics and long-term follow-up of clinical outcomes.¹⁴ In this subanalysis of the China-AF registry, we aim to investigate the following: 1) the association between small LV and CV events; and 2) whether LV size modified the efficacy of catheter ablation on reducing CV events in AF patients.

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METHODS

STUDY POPULATION. China-AF is a prospective and multicenter registry cohort of patients with AF. The rationale and design of the China-AF study have been previously described.¹⁴ In brief, patients over age 18 years with documented AF by ECG have been recruited from 20 tertiary and 12 nontertiary hospitals in Beijing, China, since August 2011. Ethics approval was obtained from the Human Research Ethics Committees at Beijing Anzhen Hospital. Institutional Review Boards in each participating hospital approved their participation. All participants in the cohort obtained written informed consent. In this analysis, we included 7,764 patients with baseline echocardiography assessment recruited between August 2011 and December 2017 (Figure 1). Follow-up surveys were conducted every 6 months at outpatient clinics or through telephone interviews until December 2020. Information on clinical outcomes, medication, and health care was collected during follow-ups.

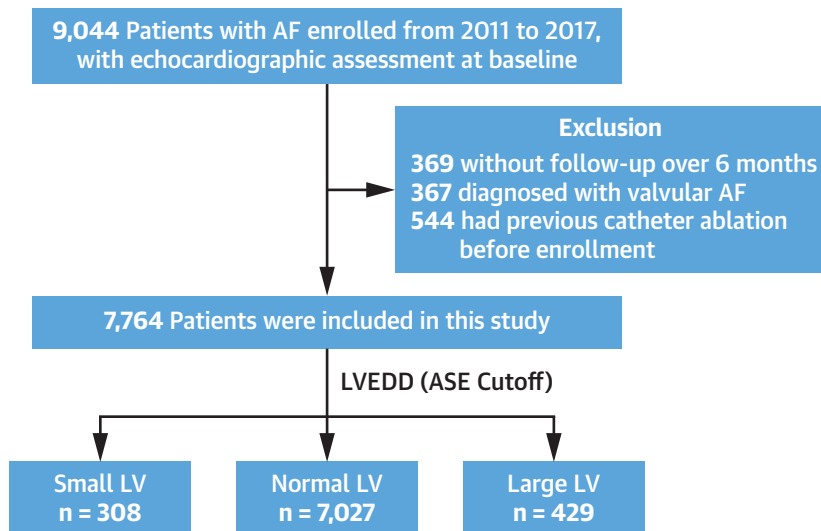
LV SIZE ASSESSMENT. Size of LV was evaluated by left ventricular end-diastolic dimension (LVEDD) measured in transthoracic echocardiography. Measurement of LVEDD was conducted according to the American Society of Echocardiography (ASE)

guidelines (Supplemental Methods).¹⁵ Normal reference values for LVEDD in ASE guidelines are 42.0 to 58.4 mm for men and 37.8 to 52.2 mm for women, which have been widely applied.¹⁵ Accordingly, small LV was defined as LVEDD <42.0 mm for men or <37.8 mm for women, and large LV was defined as LVEDD >58.4 mm for men or >52.2 mm for women.

OUTCOMES. The primary outcome of this study was a composite of CV death, ischemic stroke (IS) or systematic embolism (SE), or major bleeding events during follow-up period. Secondary outcomes were the individual components of the primary outcome. CV death included death caused by myocardial infarction, heart failure, cardiac sudden death, intracranial hemorrhage, ischemic stroke, and others. The SE event included the acute loss of perfusion of a limb or organ (visceral-mesenteric, renal, splenic arteries, and so on). Major bleeding was diagnosed according to the following: 1) transfusion of at least 2 units of whole blood or erythrocytes; 2) requiring hospitalization or surgery; 3) resulting in permanent disability; or 4) involving a critical anatomic site. More detailed definition of study endpoints can be found in the protocol of China-AF study (Supplemental Table 1).¹⁴ All reported cases of endpoints were adjudicated by an independent endpoint committee.

STATISTICAL ANALYSIS. Continuous variables were presented as median (Q1–Q3) for non-normally distributed data or mean ± SD for normal distributed data, and categorical variables were presented as percentages. Baseline characteristics among the 3 LV size groups (small LV, normal LV, and large LV) were compared using the Kruskal Wallis/1-way analysis of variance test for continuous variables or chi-square test for categorical variables. Poisson models were used to estimate the 95% CIs of incidences of clinical outcomes. Data were censored at the time when the study outcome first occurred, or the last follow-up or death occurred or maximum follow-up time of 5 years. The Kaplan-Meier estimates of survival free from primary and secondary outcomes in 3 LV size groups were calculated. The associations between LV size and the outcomes were assessed using Cox proportional regression models. The proportional hazard assumption was examined by the scaled Schoenfeld residuals test. The selection of adjusted covariate sets was determined by the minimal set using directed acyclic graph (Supplemental Figure 1). Adjusted variables included age; sex; body mass index (BMI); AF type; heart failure; previous ischemic stroke; hypertension; coronary artery disease (CAD); diabetes; chronic kidney disease; hyperlipidemia; peripheral

FIGURE 1 The Study Flow Diagram



The number of patients assessed for eligibility and the reasons for exclusion. AF = atrial fibrillation; LV = left ventricle/ventricular; LVEDD = left ventricular end-diastolic dimension.

arterial disease; catheter ablation; smoking status; drinking status; left atrial diameter; and use of oral anticoagulant, antiplatelet drugs, angiotensin-converting enzyme inhibitors/angiotensin receptor blockers, and statin (detailed definition of comorbidities in [Supplemental Table 2](#)). Individually missing values (0.4% missing BMI) for the adjusted covariates parameters were imputed using the median value (BMI 25.4 kg/m²). In the comparison of baseline characteristics, we strictly used available, nonimputed data to ensure the integrity and accuracy of these comparisons. Nonlinear associations between baseline LVEDD and outcomes were evaluated using restricted cubic splines fitted for multivariable Cox models. The reference values (HR: 1) were set at 10th percentiles with 3 knots at the 10th, 50th, and 90th of LVEDD. The number of knots was selected based on the minimal Akaike information criterion estimate. To test whether the nonlinear association between LVEDD and primary outcome was consistent across prescribed subgroups, we further conducted adjusted restricted cubic splines and tested cross-subgroup interactions in populations stratified by gender, age, BMI, catheter ablation, diabetes, and LV hypertrophy. Additionally, the HRs and 95% CIs of catheter ablation and outcomes in 3 LV size groups were also evaluated in multivariable Cox models.

A series of sensitivity analyses were conducted to validate the robustness of the results. Considering

that patients enrolled were from the Chinese Han population, we used another set of reference values for LVEDD (38.4 to 54.0 mm in men and 36.7 to 49.7 mm in women) specific to the Chinese Han population to categorize the small, large, and normal LV groups.¹⁶ Furthermore, propensity score matching was performed to balance baseline characteristics between the small and normal LV groups ([Supplemental Methods](#)). The association of small LV and outcomes in the matched cohort was also evaluated using mixed effect Cox proportional models. Additionally, the Fine-Gray model was used to assess the associations between small LV and IS/SE as well as major bleeding events, with death treated as a competing risk. Furthermore, we excluded patients with CAD (potentially concomitant with LV aneurysm or apical akinesia) in analysis to avoid possible influence of misclassification of LV size in the echocardiographic assessment. The sensitivity analysis based on the body surface area-indexed LVEDD was also performed. A 2-sided $P < 0.05$ was considered statistically significant. All analyses were conducted using R software version 4.2.0 (R Foundation for Statistical Computing).

RESULTS

BASILINE CHARACTERISTICS. Between August 2011 and December 2017, 9,044 patients were recruited from the China-AF registry, with echocardiographic

TABLE 1 Baseline Characteristics				
	Normal LV (n = 7,027)	Small LV (n = 308)	Large LV (n = 429)	P Value
Age, y	63 (55-71)	65 (57-73) ^a	64 (54-72)	<0.001
Female	2,536 (36.1)	68 (22.1) ^a	205 (47.8)	<0.001
Height, cm	170 (161-174)	170 (162-173)	167 (160-174)	0.040
Weight, kg	72 (64-80)	67.8 (60-75) ^a	73 (65-82)	<0.001
BMI, kg/m ²	25.4 (23.4-27.7)	24.2 (22.1-26.0) ^a	26 (23.7-28.8)	<0.001
SBP, mm Hg	128 (120-138)	121 (117-135) ^a	126 (119-138)	0.010
DBP, mm Hg	80 (70-84)	78 (70-81) ^a	80 (70-86)	0.039
Persistent AF	2,409 (34.3)	96 (31.2)	203 (47.3)	<0.001
Medical history				
Heart failure	478 (6.8)	26 (8.4)	174 (40.6)	<0.001
Ischemic stroke	838 (11.9)	48 (15.6)	65 (15.2)	0.027
Hypertension	4,393 (62.5)	162 (52.6) ^a	309 (72.0)	<0.001
CAD	923 (13.1)	37 (12.0)	87 (20.3)	<0.001
Diabetes	1,630 (23.2)	71 (23.1)	129 (30.1)	0.005
CKD	152 (2.2)	12 (3.9) ^a	30 (7.0)	<0.001
Hyperlipidemia	2,008 (28.6)	80 (26.0)	130 (30.3)	0.439
PAD	72 (1.0)	3 (1.0)	10 (2.3)	0.040
Echocardiography				
LA diameter, mm	40.0 (36.0-44.0)	38.0 (33.0-42.0) ^a	46.0 (41.0-50.0)	<0.001
LVEF, %	64 (60-68)	65 (61-70) ^a	52 (42-60)	<0.001
LVEDD, mm	48.0 (45.0-50.0)	40.0 (37.0-41.0) ^a	60.0 (56.0-62.0)	<0.001
LVESD, mm	31.0 (28.0-33.0)	26.0 (23.0-27.0) ^a	44.0 (37.8-48.0)	<0.001
IVS, mm	10.0 (9.0-10.4)	9.2 (8.0-11.0)	10.0 (9.0-11.0)	0.568
LVPW, mm	9.0 (8.5-10.0)	9.0 (8.0-10.0)	9.6 (8.0-10.3)	0.009
Current smoking	886 (12.6)	41 (13.3)	59 (13.8)	0.746
Current alcohol drinking	916 (13.0)	48 (15.6)	49 (11.4)	0.253
Catheter ablation	4,442 (63.2)	191 (62.0)	158 (36.8)	<0.001
Medications				
OAC	5,623 (80.0)	244 (79.2)	287 (66.9)	<0.001
Antiplatelet	953 (13.6)	51 (16.6)	115 (26.8)	<0.001
AAD	3,507 (49.9)	148 (48.1)	133 (31.0)	<0.001
β-blocker	2,731 (38.9)	122 (39.6)	268 (62.5)	<0.001
Calcium-channel blocker	226 (3.2)	6 (1.9)	15 (3.5)	0.430
Digoxin	207 (2.9)	12 (3.9)	84 (19.6)	<0.001
ACEI/ARB	2,090 (29.7)	71 (23.1) ^a	220 (51.3)	<0.001
Statin	2,650 (37.7)	110 (35.7)	176 (41.0)	0.288

Values are median (Q1-Q3) or n (%). ^aThere is statistically significant difference ($P < 0.05$) between normal left ventricle (LV) and small LV groups.

AAD = antiarrhythmic drugs; ACEI = angiotensin-converting enzyme inhibitors; AF = atrial fibrillation; ARB = angiotensin receptor blockers; BMI = body mass index; DBP = diastolic blood pressure; CAD = coronary artery disease; CKD = chronic kidney disease; IVS = interventricular septum; LA = left atrium; LVEDD = left ventricular end-diastolic dimension; LVEF = left ventricular ejection fraction; LVESD = left ventricular end-systolic dimension; LVPW = left ventricular posterior wall; OAC = oral anticoagulant; PAD = peripheral artery disease; SBP = systolic blood pressure.

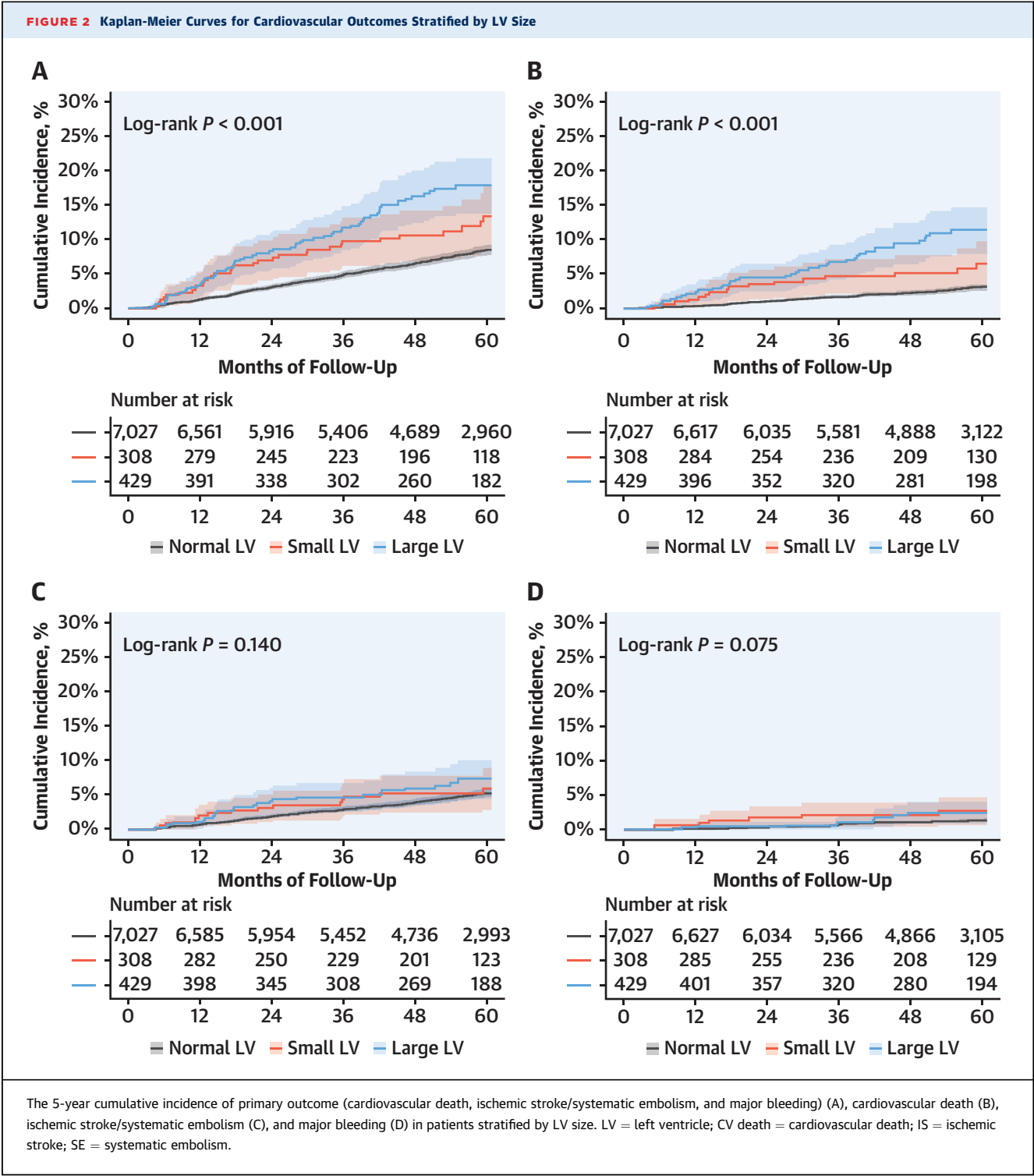
assessment at baseline. After exclusion of 1,280 ineligible patients, a total of 7,764 patients were included in final analysis (Figure 1). The median age of the 7,764 participants was 63 years (Q1-Q3: 55-71 years), with 2,809 (36.2%) women. The median LVEDD was 48.0 mm (Q1-Q3: 45.0-51.0 mm). Among them, 308 patients (4.0%) were categorized as having a small LV, 7,027 (90.5%) as normal LV, and 429 (5.5%) as large LV. Patients with small LV, compared with those with a normal LV, were more likely to carry a lower BMI, lower blood pressure, normal left

ventricular ejection fraction (LVEF), and fewer comorbidities (Table 1). Reduced diameters of both atriums and right ventricle were observed in small LV patients, and there was less prevalence of pulmonary hypertension assessed by echocardiography in the small LV group (Supplemental Table 3). The level of hemoglobin and red blood cell counts was higher among patients with small LV; level of B-type natriuretic peptide and high-sensitivity C-reactive protein were similar between normal LV and small LV groups (Supplemental Table 4). No significant differences in the type of anticoagulation used and doses administered were observed between normal LV and small LV subgroups (Supplemental Table 5). By contrast, patients with large LV had a higher proportion of obesity, reduced LVEF, more comorbidities, and a lower proportion of catheter ablation (Table 1). The prevalence of heart failure was highest in the large LV group (40.6% overall HF, 18.9% HF with reduced ejection fraction), followed by the small LV group (8.4% overall HF, 7.14% HF with preserved ejection fraction) (Supplemental Figure 2). A higher level of B-type natriuretic peptide and high-sensitivity C-reactive protein was observed in the large LV group (Supplemental Table 4).

ASSOCIATIONS BETWEEN SMALL LV AND CV OUTCOMES.

During a median follow-up time of 1,639 days (Q1-Q3: 1,102-1,825 days), the primary composite outcome occurred in 572 patients (7.4%), with 224 CV deaths, 330 IS/SE, and 95 major bleeding events, respectively.

Incidence of the primary outcome was higher in both small LV (29.06; 95% CI: 20.01-40.83 per 1,000 person-years) and large LV (40.85; 95% CI: 31.45-52.15 per 1,000 person-years) groups comparing with the normal LV group (17.57; 95% CI: 16.03-19.23 per 1,000 person-years) (Figure 2, Table 2). After adjusting for clinical confounding variables and echocardiographic parameters, small LV was independently associated with an increased risk of the primary outcome (adjusted HR: 1.54; 95% CI: 1.07-2.20; $P = 0.019$) (Table 2). A significant association of large LV with the primary outcome was also observed (adjusted HR: 1.36; 95% CI: 1.02-1.81; $P = 0.033$) when accounting for clinical and echocardiographic confounders (Table 2). Adjusted restricted cubic splines fitted for multivariable Cox proportional model indicated a U-shaped association between LVEDD and the primary outcome, indicating that the risk of primary outcome increased with the decrease in LVEDD below the 10th percentile of LVEDD ($P_{\text{overall}} = 0.001$; $P_{\text{nonlinear}} < 0.001$) (Figure 3). The nonlinear associations between LVEDD and the risk of the primary outcome were persistent among subgroups stratified by gender, age,



BMI, catheter ablation, diabetes, and LV hypertrophy (Figure 4).

Patients with small LV were significantly associated with an increased risk of CV death (adjusted HR:1.94; 95% CI: 1.14-3.28; $P = 0.014$) (Figure 2, Table 2). The relationship of LVEDD with CV death and major bleeding also presented U-shaped associations (Figure 3). Heart failure was the leading CV

TABLE 2 The Association Between LV Size and Cardiovascular Outcomes						
	Events/Person-y	Incidence Rate (95%CI), per 1,000 Person-y	Unadjusted HR (95%CI)	P Value	Adjusted HR (95%CI)	P Value
Primary outcome ^a						
Normal LV	475/27,031	17.57 (16.03-19.23)	Ref		Ref	
Small LV	33/1,135	29.06 (20.01-40.83)	1.66 (1.17-2.36)	0.005	1.54 (1.07-2.20)	0.019
Large LV	64/1,567	40.85 (31.45-52.15)	2.33 (1.80-3.03)	<0.001	1.36 (1.02-1.81)	0.033
CV death						
Normal LV	168/27,642	6.08 (5.19-7.07)	Ref		Ref	
Small LV	16/1,180	13.55 (7.75-22.02)	2.24 (1.34-3.73)	0.002	1.94 (1.14-3.28)	0.014
Large LV	40/1,631	24.53 (17.52-33.4)	4.04 (2.86-5.71)	<0.001	1.83 (1.24-2.69)	0.002
IS/SE						
Normal LV	290/27,197	10.66 (9.47-11.96)	Ref		Ref	
Small LV	15/1,155	12.98 (7.27-21.42)	1.22 (0.73-2.05)	0.454	1.18 (0.69-1.99)	0.548
Large LV	25/1,596	15.67 (10.14-23.12)	1.47 (0.98-2.21)	0.064	1.02 (0.66-1.57)	0.945
Major bleeding						
Normal LV	80/27,608	2.90 (2.30-3.61)	Ref		Ref	
Small LV	7/1,177	5.95 (2.39-12.25)	2.06 (0.95-4.46)	0.067	2.21 (1.01-4.86)	0.048
Large LV	8/1,638	4.88 (2.11-9.62)	1.70 (0.82-3.51)	0.153	1.16 (0.53-2.53)	0.707
^a The primary outcome was the composite of incident CV death, IS/SE, or major bleeding. IS = ischemic stroke; SE = systematic embolism; other abbreviations as in Table 1.						

cause of death across LV size groups, with a notably higher proportion observed in the small LV group compared with those with normal LV dimension; among patients with small LV, myocardial infarction and intracranial hemorrhage equally ranked as the secondary CV causes of death (Supplemental Table 6).

Small LV was also significantly associated with an increased risk of major bleeding (adjusted HR: 2.21; 95% CI: 1.01-4.86; $P = 0.048$) (Figure 2, Table 2), and the association between anticoagulation and major bleeding was similar between the small and normal LV groups (Supplemental Table 7). However, small LV was not observed to be significantly associated with increased risk of IS/SE events (adjusted HR: 1.18; 95% CI: 0.69-1.99; $P = 0.548$) (Figure 2, Table 2).

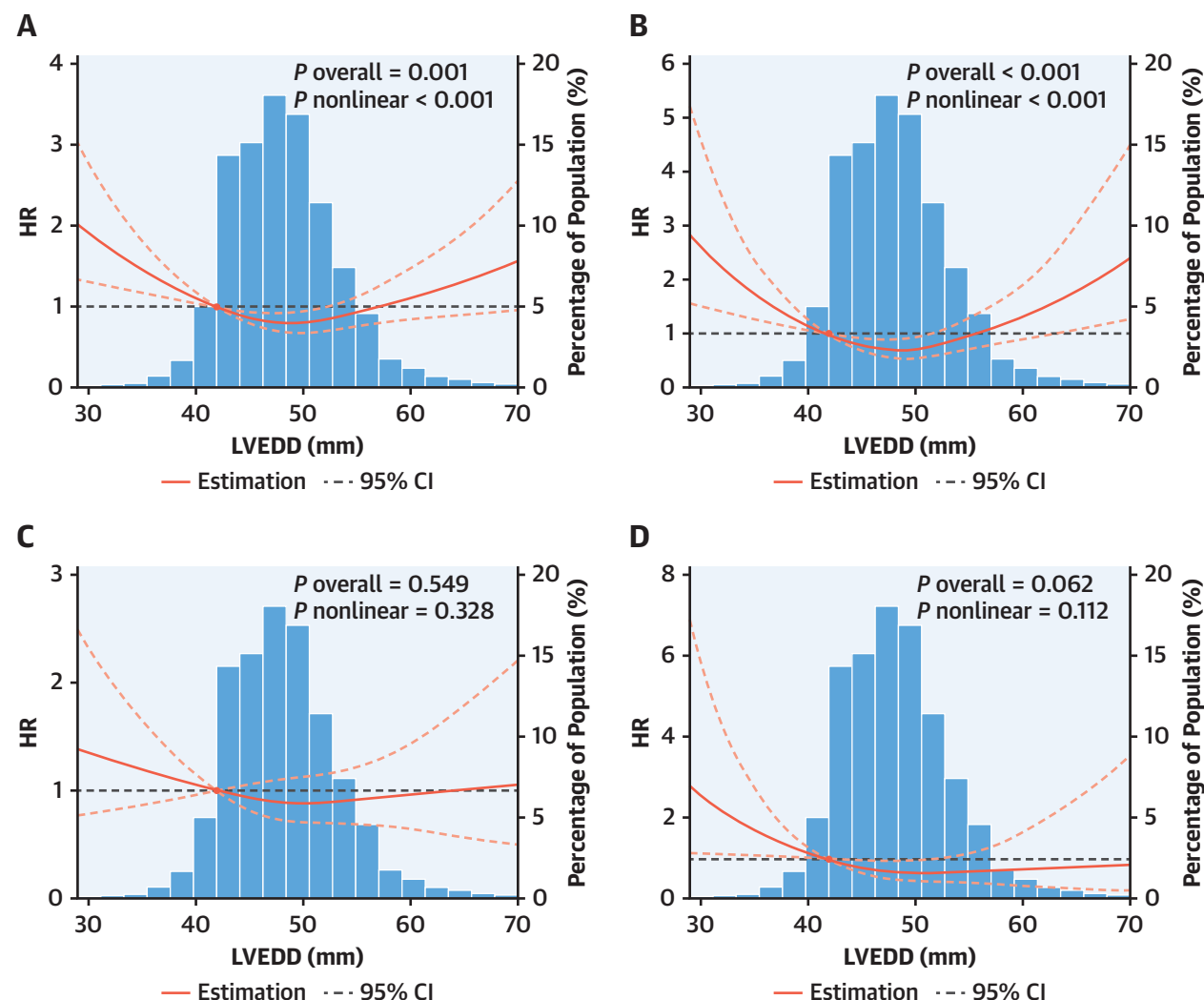
EFFECTS OF CATHETER ABLATION ON CV OUTCOMES ACROSS LV SIZE GROUPS. For the primary outcome, adjusted HR of catheter ablation was 0.77 (95% CI: 0.31-1.90) in the small LV group, 0.62 (95% CI: 0.49-0.78) in the normal LV group, and 0.23 (95% CI: 0.09-0.63) in the large LV group ($P_{\text{for interaction}} = 0.152$) (Figure 5). There were also no statistically significant interactions between LV size and the effect of catheter ablation on CV death and major bleeding (Figure 5). However, the probably protective effect of catheter ablation on reducing IS/SE event incidence was stronger (adjusted HR: 0.13; 95% CI: 0.03-0.61; $P = 0.010$) in the large LV group than small LV group (adjusted HR: 2.18; 95% CI: 0.52-9.20; $P = 0.290$) and normal LV group (adjusted HR: 0.76; 95% CI: 0.57-1.02; $P = 0.065$) ($P_{\text{for interaction}} = 0.020$) (Figure 5).

SENSITIVITY ANALYSIS. In the sensitivity analysis using LVEDD cutoffs for the Chinese Han population, 95 patients were categorized into the small LV group, 6,594 patients in the normal LV group, and 1,075 patients in the large LV group. Patients with small LV consistently showed a higher risk of the primary composite outcome, CV death, and major bleeding (Supplemental Table 8). Still, small LV remained not associated with incident IS/SE events. In a 1:2 propensity score-matched cohort with balanced baseline characteristics (Supplemental Table 9), the significant associations between small LV and an increased risk of the primary CV outcome and CV death persisted (Supplemental Table 10). When taking into account the competing risk of death, small LV remained significantly associated with an increased risk of major bleeding (Supplemental Table 11). The results in the sensitivity analysis using body surface area-indexed LVEDD suggested the similar nonlinear association with adverse outcomes (Supplemental Figure 3). After excluding patients with CAD, the nonlinear association between LVEDD and worse prognosis was still observed (Supplemental Figure 4).

DISCUSSION

To the best of our knowledge, the present study is the first to identify that, in patients with AF, small LV was independently associated with an increased risk of the composite outcome of CV death, IS/SE, or major bleeding (Central Illustration). Additionally, the

FIGURE 3 Adjusted Association of LVEDD and Cardiovascular Outcomes



Adjusted associations of measures of left ventricular end-diastolic dimension (LVEDD) and incident primary outcome (cardiovascular death, ischemic stroke/systematic embolism, or major bleeding) (A), cardiovascular death (B), ischemic stroke/systematic embolism (C), and major bleeding (D). The reference value (HR: 1) was set at the 10th percentile of the LVEDD (42 mm). The solid lines represent the adjusted HRs of LVEDD across the whole range and the dotted lines represent the corresponding 95% CIs.

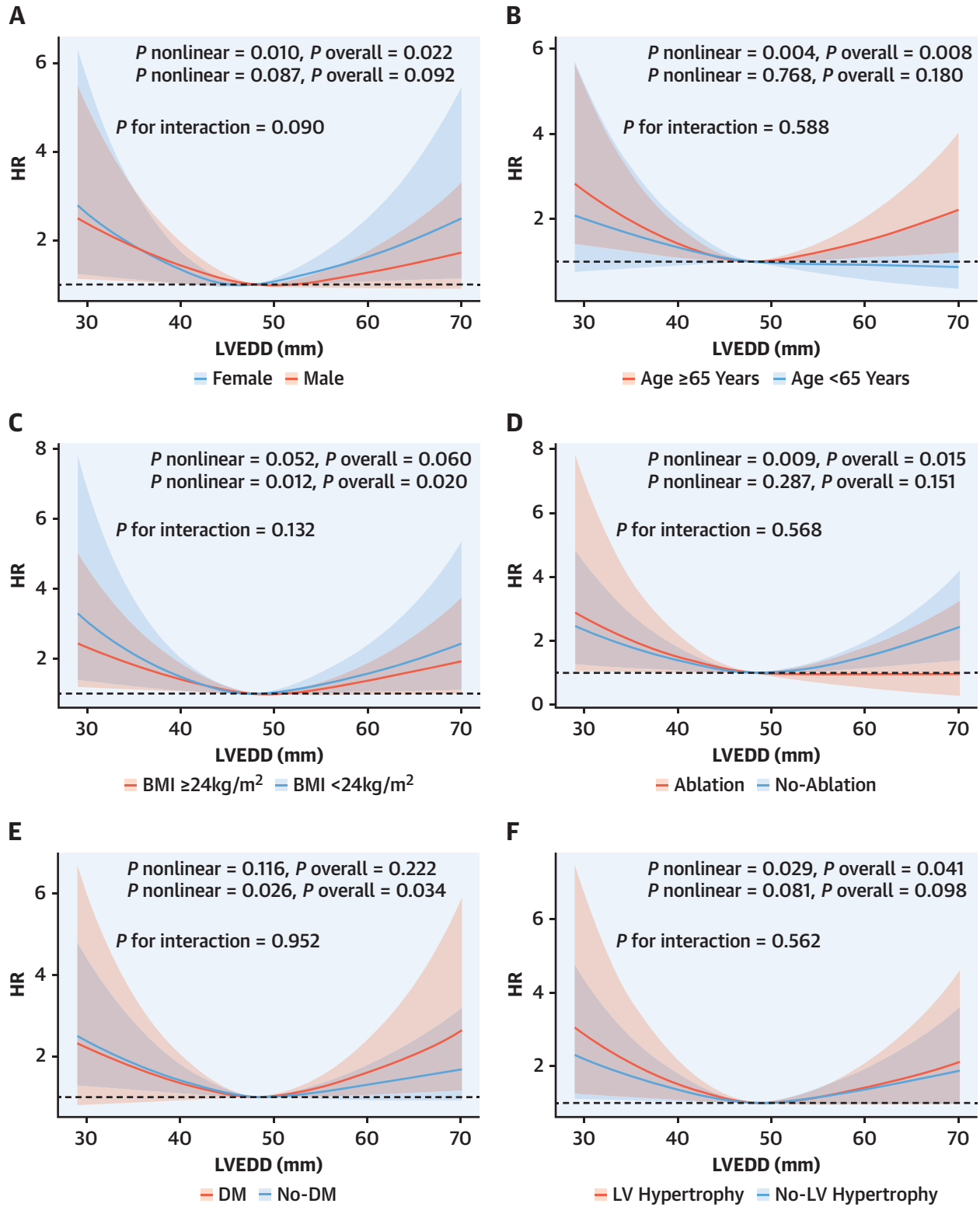
beneficial effect of catheter ablation in reducing primary CV events remained regardless of LV size.

The investigation into the small LV phenotype was initially sparked by observations in patients with chronic fatigue syndrome, who were found to have reduced LV diameters and stroke volume.^{7,17} Impaired oxygen transfer, energy metabolism, and hemodynamic regulation seem to be the main offenders of the correlation between circulatory volume deficit and the severity of fatigue.⁸ In the present study, all heart chamber sizes were lower in patients

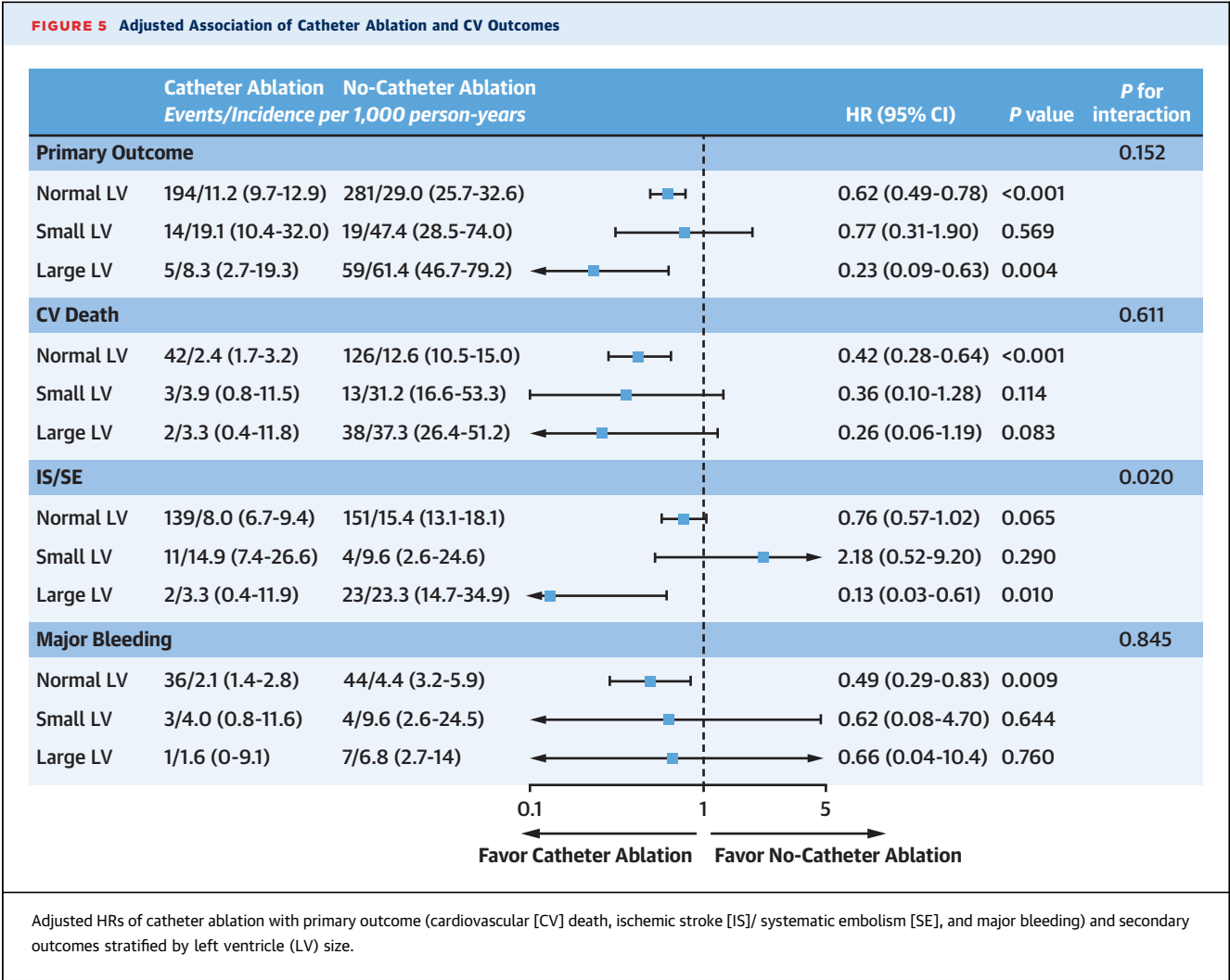
with small LV, reflecting a shrink in the overall cardiac size. In the small-LV population, lower blood pressure and a diminished likelihood of pulmonary hypertension were noted, alongside an increase in hemoglobin and red blood cell counts, in contrast to the normal LV group. Although lacking a direct hemodynamic assessment, these clinical manifestations may hint at the potential for a low-output circulatory state in patients with small LV.

Leveraging the largest AF registry database in China, our study observed that small LV was

FIGURE 4 Adjusted Association of LVEDD and Primary Outcome in Subgroups



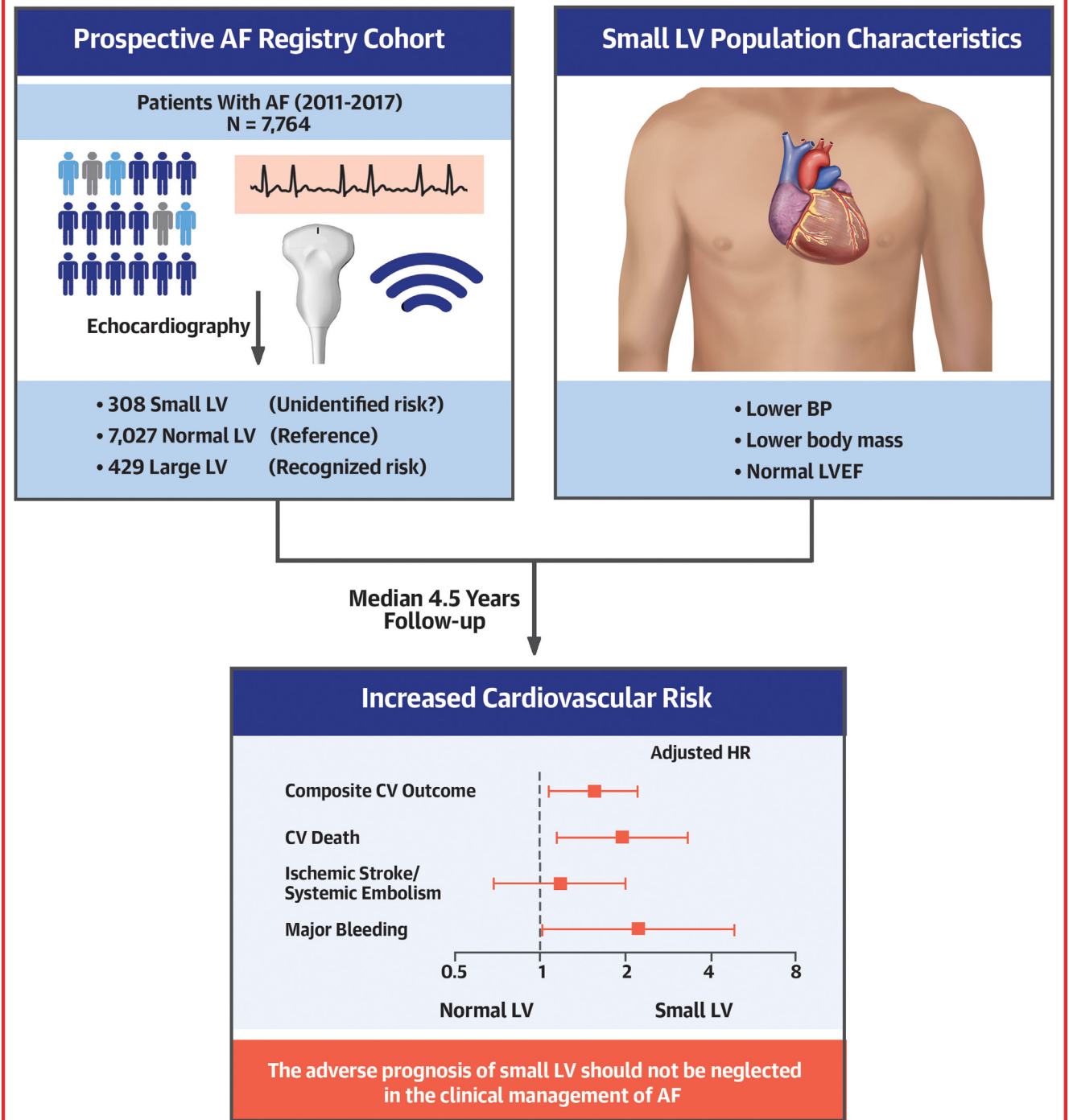
(A-F) Adjusted associations of measures of left ventricular end-diastolic dimension (LVEDD) and incident primary outcome (cardiovascular death, ischemic stroke/systematic embolism, or major bleeding) in subgroups. The reference values (HR: 1) were set at the median value of each subgroup. The solid lines represented the adjusted HRs of LVEDD across the whole range and the shaded areas represented the corresponding 95% CIs.



associated with increased risk of CV mortality and major bleeding, independent of demographic and clinical characteristics. Several potential explanations may elucidate the association between small LV and worse clinical outcome in AF patients. First, during the exacerbation of AF, impaired atrial contraction contributed to a reduction of ventricular filling.¹³ A smaller heart necessitated hypercontract efforts to generate the required cardiac output, resulting in excessive cardiac metabolic burden potentially contributing to an unfavorable prognosis.¹⁸ The elevated LVEF observed in this study was consistent with previous observations, indicating a compensatory response to counteract the stroke volume deficit.⁸ Second, small LV seems to exhibit poorer tolerance to changes in volume, rendering it more vulnerable to alterations in preload. The OCEAN-TAVI (Optimized Transcatheter Valvular Intervention-Transcatheter Aortic Valve Implantation) registry reported that, in

patients who developed paravalvular leak following transcatheter aortic valve replacement, those with small LV exhibited a higher risk of death and rehospitalization caused by heart failure.¹⁰ Third, volume depletion-relevant neurohumoral regulatory mechanism changes, such as renin-angiotensin-aldosterone system activation and autonomic nervous system dysregulation, may also play a role in the pathogenesis and progression of atrial arrhythmia and downstream clinical outcomes.^{19,20} Besides, the genetic factors involving in the development of cardiac chamber may also contribute to the elevated CV risk in small LV patients. A recent Mendelian randomization study reported that the genetic variants predicting low birth weight were associated with small chamber volumes and lower stroke volume, and carried a higher risk of CV disease.²¹ It has been increasingly recognized that AF adversely affects cardiac performance, through loss

CENTRAL ILLUSTRATION Association Between Small LV and Increased Cardiovascular Risk in AF



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In a prospective AF cohort, patients with small LV presented unique clinical manifestations, and were associated with an increased cardiovascular risk in the long-term follow-up. AF = atrial fibrillation; BP = blood pressure; CV = cardiovascular; IS = ischemic stroke; LV = left ventricle/ventricular; LVEF = left ventricular ejection fraction; SE = systematic embolism.

of atrial contraction, tachycardia, neurohumoral activation, and structural myocardial changes.^{22,23} In our study, heart failure was the leading CV cause of death across all subgroups, which was consistent with the current data from AF registries.²⁴ A hemodynamic study revealed significant impairment of LV systolic and diastolic functions in simulated AF attack, which could pose a threat to the prognosis of small-LV patients.²² Theoretical simulation suggested that a small heart with a maximal stroke volume of 60 mL only generates a cardiac output of 9 L/min with 150 beats/min of heart rate, implying that the mechanically inefficiency of the small LV may hinder its capacity to deal with hemodynamic challenges.⁸ Therefore, intensifying follow-up on the conditions of this fragile population may facilitate a deeper comprehension of the disease progression.

In our study, AF patients with small LV showed higher major bleeding risk in the long-term, despite with no significant differences in the antithrombotic regimens compared with patients with normal LV. It is worth noting that the increased risk of CV death and major bleeding may also be ascribed to certain body degeneration traits, such as frailty and older age.²⁵ In the current analysis, it was also observed that patients with small LV tended to exhibit lower body mass and older age than those with a normal LV size, implying a potential role of physical constitution in the predisposition to adverse outcomes. We found that most major bleeding events occurred in the small-LV patients receiving warfarin treatment. The drug concentration may be higher in patients with lower body weight, raising a challenge to adjusting the warfarin dose for safely achieving the target international normalized ratio range.²⁶ It should be noted that most patients receiving direct oral anticoagulant (DOAC) treatment were prescribed low-dose DOAC anticoagulation, reflecting the cautious approach to DOAC dosing caused by bleeding concerns in Asia.²⁷ However, the safety of standard-dose DOAC is still unclear caused by the small sample size in this treatment group, warranting further investigation to evaluate whether standard DOAC anticoagulation is suitable for such patients.^{28,29}

Our analysis did not detect a significant interaction effect between catheter ablation and LV size in CV death and major bleeding events. Hence, performing catheter ablation in patients with small LV might be plausible, considering its potential benefits of restoring cardiac volume output.³⁰ A recent randomized trial reported sinus rhythm restoration through catheter ablation resulted in significant improvements in cardiac output and exercise capacity in HF patients, which was also supported by previous

observational studies.^{31,32} In small-LV patients, catheter ablation rescued volume hypoperfusion aggravation caused by AF, which might explain the consistent beneficial effect in reducing mortality. Interestingly, we also found that in patients with large LV, receiving catheter ablation was associated with more pronounced reduction of embolism. In our study population, enlarged left atrium and persistent AF were common in large-LV patients, suggesting a high AF burden with elevated risk of embolism.

In this large real-world cohort, we have demonstrated a U-shaped association between LV size and adverse CV events. The finding challenges traditional opinions that only LV enlargement matters. Actually, small LV is not rare in clinical practice but did not receive much attention. It remains unknown whether the negative effect of small LV is solely a consequence of chamber size decrease accompanied by hemodynamic changes or merely one of imaging manifestations of the degenerative condition. Nonetheless, considering the significant association of small LV with adverse outcomes, it might be plausible to consider small LV as a new structural risk factor in predicting the prognosis of patients with AF.

STUDY LIMITATIONS. As a subanalysis of a large-scale and multicenter registry, this study has inherent defects of observational research. Although multiple adjustments for demographic and clinical characteristics were adopted in analysis, residential confounding and unmeasured confounding would remain. However, identified associations still merit attention to the potentially worse prognosis of small-LV patients. The underlying cause of higher CV risk among AF patients with a small LV is complex and may not be thoroughly explained by this study. Nonetheless, fewer comorbidities were present in patients with a small LV, suggesting unique mechanisms beyond other concomitant structural influences. Although we utilized the widely applied echocardiography method to assess LV size, misclassification of LV size in echocardiography may occur in patients with LV aneurysm or apical akinesia. It may be noted that there was small proportion of previous CAD in this study population. Besides, the nonlinear association between LVEDD and prognosis persisted when excluding patients with CAD. As the gold standard for cardiac structure and function, cardiac magnetic imaging has not been widely performed in the registry sites at the baseline period, which limited a direct validation on the echocardiographic parameters. The comprehensive data of LV diastolic function, right ventricular function, and stroke volume were also unavailable in the registry database, caused

by their infrequent measurement in clinical practice. However, we extracted related echocardiographic parameters in the registry database, which could partially reflect the cardiac conditions in the population. In the future, an elaborate cardiac hemodynamic assessment may be helpful in exploring the mechanism of poor prognosis in small LV population. Last, this study was conducted in a Chinese population, and it remains uncertain whether small LV is also a risk factor in other ethnic groups.

CONCLUSIONS

In this analysis of a large multicenter registry of patients with AF, small LV was associated with an increased risk of the composite outcome of CV death, embolism, and major bleeding. Identification of small LV as an important anatomic risk factor may be needed in the risk stratification and management of patients with AF.

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PERSPECTIVES

COMPETENCY IN PATIENT CARE AND

PROCEDURAL SKILLS: Patients with AF and a small LV are at increased risk of adverse CV events.

TRANSLATIONAL OUTLOOK: Additional research is needed to identify the pathophysiological mechanisms linking small LVs with adverse clinical outcomes.

REFERENCES

- Odutayo A, Wong CX, Hsiao AJ, Hopewell S, Altman DG, Emdin CA. Atrial fibrillation and risks of cardiovascular disease, renal disease, and death: systematic review and meta-analysis. *BMJ*. 2016;354:i4482.
- Du X, Guo L, Xia S, et al. Atrial fibrillation prevalence, awareness and management in a nationwide survey of adults in China. *Heart*. 2021;107:535–541.
- Turagam MK, Musikantow D, Whang W, et al. Assessment of catheter ablation or antiarrhythmic drugs for first-line therapy of atrial fibrillation: a meta-analysis of randomized clinical trials. *JAMA Cardiol*. 2021;6:697–705.
- Donal E, Lip GY, Galderisi M, et al. EACVI/EHRA expert consensus document on the role of multimodality imaging for the evaluation of patients with atrial fibrillation. *Eur Heart J Cardiovasc Imaging*. 2016;17:355–383.
- Joglar JA, Chung MK, Armbruster AL, et al. 2023 ACC/AHA/ACCP/HRS guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2024;83:109–279.
- Zile MR, Gaasch WH, Patel K, Aban IB, Ahmed A. Adverse left ventricular remodeling in community-dwelling older adults predicts incident heart failure and mortality. *J Am Coll Cardiol HF*. 2014;2:512–522.
- Miwa K. Cardiac dysfunction and orthostatic intolerance in patients with myalgic encephalomyelitis and a small left ventricle. *Heart Vessels*. 2015;30:484–489.
- Natelson BH, Brunjes DL, Mancini D. Chronic fatigue syndrome and cardiovascular disease: JACC state-of-the-art review. *J Am Coll Cardiol*. 2021;78:1056–1067.
- Alkady H, Saber A, Abouramadan S, Elnaggar A, Nasr S, Mahmoud E. Mitral valve replacement in mitral stenosis; the problem of small left ventricle. *J Cardiothorac Surg*. 2020;15:67.
- Saito T, Inohara T, Yoshijima N, et al. Small Left Ventricle and Clinical Outcomes After Transcatheter Aortic Valve Replacement. *J Am Heart Assoc*. 2021;10:e019543.
- Kawabori M, Kurihara C, Sugiura T, et al. Continuous-flow left ventricular assist device implantation in patients with a small left ventricle. *Ann Thorac Surg*. 2018;105:799–806.
- Pagliaro BR, Mincione G, Taormina A, et al. Atrial arrhythmias and heart failure: A "modern view" of an old paradox. *Pacing Clin Electrophysiol*. 2023;46:395–408.
- Reddy YNV, Borlaug BA, Gersh BJ. Management of atrial fibrillation across the spectrum of heart failure with preserved and reduced ejection fraction. *Circulation*. 2022;146:339–357.
- Du X, Ma C, Wu J, et al. Rationale and design of the Chinese Atrial Fibrillation Registry Study. *BMC Cardiovasc Disord*. 2016;16:130.
- Lang RM, Badano LP, Mor-Avi V, et al. Recommendations for cardiac chamber quantification by echocardiography in adults: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *J Am Soc Echocardiogr*. 2015;28:1–39.e14.
- Yao GH, Zhang M, Yin LX, et al. Doppler Echocardiographic Measurements in Normal Chinese Adults (EMINCA): a prospective, nationwide, and multicentre study. *Eur Heart J Cardiovasc Imaging*. 2016;17:512–522.
- Olimulder MA, Galjee MA, Wagenaar LJ, et al. Chronic fatigue syndrome in women assessed with combined cardiac magnetic resonance imaging. *Neth Heart J*. 2016;24:709–716.

18. Kerkhof PLM, Kuznetsova T, Ali R, Handly N. Left ventricular volume analysis as a basic tool to describe cardiac function. *Adv Physiol Educ*. 2018;42:130-139.
19. Mascolo A, Urbanek K, De Angelis A, et al. Angiotensin II and angiotensin 1-7: which is their role in atrial fibrillation? *Heart Fail Rev*. 2020;25:367-380.
20. Chen PS, Chen LS, Fishbein MC, Lin SF, Nattel S. Role of the autonomic nervous system in atrial fibrillation: pathophysiology and therapy. *Circ Res*. 2014;114:1500-1515.
21. Ardissino M, Morley AP, Slob EAW, et al. Birth weight influences cardiac structure, function, and disease risk: evidence of a causal association. *Eur Heart J*. 2024;45:443-454.
22. Stojadinović P, Deshraj A, Wichterle D, et al. The hemodynamic effect of simulated atrial fibrillation on left ventricular function. *J Cardiovasc Electrophysiol*. 2022;33:2569-2577.
23. Verhaert DVM, Brunner-La Rocca HP, van Veldhuisen DJ, Vernooy K. The bidirectional interaction between atrial fibrillation and heart failure: consequences for the management of both diseases. *Europace*. 2021;23:ii40-ii45.
24. Healey JS, Oldgren J, Ezekowitz M, et al. Occurrence of death and stroke in patients in 47 countries 1 year after presenting with atrial fibrillation: a cohort study. *Lancet*. 2016;388:1161-1169.
25. Ijaz N, Buta B, Xue QL, et al. Interventions for frailty among older adults with cardiovascular disease: JACC state-of-the-art review. *J Am Coll Cardiol*. 2022;79:482-503.
26. Hart R, Veenstra DL, Boudreau DM, Roth JA. Impact of body mass index and genetics on warfarin major bleeding outcomes in a community setting. *Am J Med*. 2017;130:222-228.
27. Chan YH, Kuo CT, Yeh YH, et al. Thromboembolic, bleeding, and mortality risks of rivaroxaban and dabigatran in Asians with nonvalvular atrial fibrillation. *J Am Coll Cardiol*. 2016;68:1389-1401.
28. Volgman AS, Nair G, Lyubarova R, et al. Management of atrial fibrillation in patients 75 years and older: JACC state-of-the-art review. *J Am Coll Cardiol*. 2022;79:166-179.
29. Lee SR, Choi EK, Park CS, et al. Direct oral anticoagulants in patients with nonvalvular atrial fibrillation and low body weight. *J Am Coll Cardiol*. 2019;73:919-931.
30. Turagam MK, Garg J, Whang W, et al. Catheter ablation of atrial fibrillation in patients with heart failure: a meta-analysis of randomized controlled trials. *Ann Intern Med*. 2019;170:41-50.
31. Chieng D, Sugumar H, Segal L, et al. Atrial fibrillation ablation for heart failure with preserved ejection fraction: a randomized controlled trial. *J Am Coll Cardiol HF*. 2023;11:646-658.
32. Sugumar H, Nanayakkara S, Vizi D, et al. A prospective Study using invasive haemodynamic measurements following catheter ablation for AF and early HFpEF: STALL AF-HFpEF. *Eur J Heart Fail*. 2021;23:785-796.

KEY WORDS atrial fibrillation, cardiovascular risk, catheter ablation, left ventricle, prospective cohort

APPENDIX For an expanded Methods section as well as supplemental tables and figures, please see the online version of this paper.